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Radiation

This chapter describes the radiation models used in Sentaurus Device.

When high-energy particles penetrate a semiconductor device, they deposit their energy by the generation of electron–hole pairs. These charges can perturb the normal operation of the device. This chapter describes the models for carrier generation by gamma radiation, alpha particles, and heavy ions.

- Carrier Generation by Gamma Radiation
- Carrier Generation by Alpha Particles
- Carrier Generation by Heavy Ions
- References

Carrier Generation by Gamma Radiation

To activate the gamma radiation model, specify the keyword `Radiation(...)`, with optional parameters, in the `Physics` section. For example:

```
Physics { ...
   Radiation(
      Dose = <float> | DoseRate = <float>
      DoseTime = (<float>,<float>)
      DoseTSigma = <float>
   )
}
```

where:

- `DoseRate` (in rad/s) represents *D* in Equation 812.
- The optional `DoseTime` (in s) allows you to specify the time period during which exposure to the constant `DoseRate` occurs.

- `DoseTSigma` (in s) can be combined with `DoseTime` to specify the standard deviation of a Gaussian rise and fall of the radiation exposure.
- As an alternative to `DoseRate`, you can specify `Dose` (in rad) to represent the total radiation exposure over the `DoseTime` interval. In this case, `DoseTime` must be specified.

To plot the generation rate due to gamma radiation, specify `RadiationGeneration` in the `Plot` section.

Yield Function

Generation of electron–hole pairs due to radiation is an electric field–dependent process [1] and is modeled as follows:

Equation 812

$$G_r = g_0 D \cdot Y(F)$$

$$Y(F) = \left(\frac{F+E_0}{F+E_1}\right)^m$$

where *D* is the dose rate, g_0 is the generation rate of electron–hole pairs, and E_0, E_1, and m are constants.

All these constants can be specified in the parameter file of Sentaurus Device as follows:

```
Radiation {
   g = 7.6000e+12          # [1/(rad*cm^3)]
   E0 = 0.1                # [V/cm]
   E1 = 1.3500e+06         # [V/cm]
   m = 0.9                 # [1]
}
```

Carrier Generation by Alpha Particles

To activate the alpha particle model, specify the keyword `AlphaParticle(...)`, with optional parameters, in the `Physics` section. For example:

```
Physics { ...
   AlphaParticle (<keyword_options>)
}
```

Table 293 on page 1813 lists the keyword options for alpha particles.

To plot the instant generation rate $G(t)$ due to alpha particles, specify `AlphaGeneration` in the `Plot` section. To plot $\int_{-\infty}^{\infty} dt G$, specify `AlphaCharge`.

Note:

The generation by alpha particles cannot be used except in transient simulations. The number of electron–hole pairs generated before the initial time of the transient is added to the carrier densities at the beginning of the simulation.

An option to improve the spatial integration of the charge generation is presented in References on page 868.

Using the Alpha Particle Model

The generation rate caused by an alpha particle with energy *E* is computed by [2]:

Equation 813

$$G(u, v, w, t) = \frac{a}{\sqrt{2\pi} \cdot s} \exp\left[-\frac{1}{2}\left(\frac{t - t_{\mathrm{m}}}{s}\right)^2 - \frac{1}{2}\left(\frac{v^2 + w^2}{w_{\mathrm{t}}^2}\right)\right]\left[c_1 e^{\alpha u} + c_2 \exp\left(-\frac{1}{2}\left(\frac{u - \alpha_1}{\alpha_2}\right)^2\right)\right]$$

if $u < \alpha_1 + \alpha_3$, and by:

Equation 814

$$G(u, v, w, t) = 0$$

if $u \geq \alpha_1 + \alpha_3$. In this case, u is the coordinate along the particle path, and v and w are coordinates orthogonal to u. The direction and place of incidence are defined in the `Physics` section of the command file with the keywords `Direction` and `Location` (or `StartPoint`, see #unique_959_Connect_42_i1990830), respectively. The t_{m} parameter is the time of the generation peak defined by the keyword `Time`. A Gaussian time dependence can also be used to simulate the typical generation due to pulsed laser or electron beams.

The maximum of the Bragg peak, α_1, is fitted to data [3] by a polynomial function:

Equation 815

$$\alpha_1 = a_0 + a_1 E + a_2 E^2$$

The parameter c_1 is given by:

Equation 816

$$c_1 = \exp\left[\alpha\left(\alpha_1[10\mathrm{MeV}] - \alpha_1[E]\right)\right]$$

The scaling factor a is determined from:

Equation 817

$$\int_0^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} G(u, v, w, t)\, dt dw dv du = \frac{E}{E_p}$$

where E_p is the average energy needed to create an electron–hole pair. The remaining parameters are listed in Table 143. They are available in the parameter set `AlphaParticle` and are valid for alpha particles with energies between 1 MeV and 10 MeV.

Table 143 Coefficients for carrier generation by alpha particles

Symbol	Parameter name	Default	Unit
s	`s`	2×10^{-12}	s
w_t	`wt`	1×10^{-5}	cm
c_2	`c2`	1.4	1
a	`alpha`	90	cm^{-1}
α_2	`alpha2`	5.5×10^{-4}	cm
α_3	`alpha3`	2×10^{-4}	cm
E_p	`Ep`	3.6	eV
a_o	`a0`	-1.033×10^{-4}	cm
a_1	`a1`	2.7×10^{-10}	cm/eV
a_2	`a2`	4.33×10^{-17}	cm/eV^2

Note:

For device simulations with cylindrical symmetry (see Cylindrical Coordinate System on page 61), the track center specified by `Location` and `Direction` must align with the cylindrical axis. Moreover, the `Direction` must be (0,1) only in traditional coordinates (see Cylindrical Coordinate System) to preserve cylindrical symmetry.

Carrier Generation by Heavy Ions

When a heavy ion penetrates a device structure, it loses energy and creates a trail of electron–hole pairs. These additional electrons and holes might cause a large enough

current to switch the logic state of a device, for example, a memory cell. Important factors are:

- The energy and type of the ion
- The angle of penetration of the ion
- The relation between the lost energy or linear energy transfer (LET) and the number of pairs created

The simulation of a single event upset caused by a heavy ion impact is activated by using the keyword `HeavyIon` in an appropriate `Physics` section. For example:

```
# Default heavy ion
Physics {
   HeavyIon (<keyword_options>)
}
# User-defined heavy ion
Physics {
   HeavyIon (<IonName>) (<keyword_options>)
}
```

Note:

User-defined heavy ions do not have default parameters. Therefore, in the parameter file, a specification of the following form must appear (see Table 145 on page 866):

```
HeavyIon(<IonName>)(...)
```

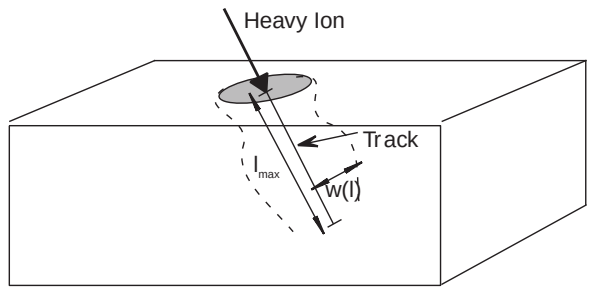
Table 294 on page 1814 describes the options for `HeavyIon`. The generation rate by the heavy ion is generally used in transient simulations. The number of electron–hole pairs generated before the initial time of the transient is added to the carrier densities at the beginning of the simulation. The total charge density and the instant generation rate are plotted using `HeavyIonChargeDensity` and `HeavyIonGeneration` in the `Plot` section, respectively.

If the value of `Wt_hi` is 0, then uniform generation is selected. If the value of `LET_f` is 0, then the keyword `LET_f` can be ignored.

Using the Heavy Ion Model

Figure 48 shows a simple model for the heavy ion impinging process.

Figure 48 A heavy ion penetrating a semiconductor; its track is defined by a length and the transverse spatial influence is assumed to be symmetric about the track axis



The generation rate caused by the heavy ion is computed by:

Equation 818

$$G(l, w, t) = G_{\mathrm{LET}}(l) R(w, l) T(t)$$

if $l < l_{\max}$ ($l_{\max}$ is the length of the track), and by:

Equation 819

$$G(l, w, t) = 0$$

if $l \geq l_{\max}$. $R(w)$ and $T(t)$ are functions describing the spatial and temporal variations of the generation rate. $G_{\mathrm{LET}}(l)$ is the LET generation density and its unit is pairs/cm^3.

$T(t)$ is defined as a Gaussian function:

Equation 820

$$T(t) = \frac{2 \cdot \exp\left(-\left(\frac{t-t_0}{\sqrt{2} \cdot s_{\mathrm{hi}}}\right)^2\right)}{\sqrt{2} \cdot s_{\mathrm{hi}} \sqrt{\pi} \left(1 + \mathrm{erf}\left(\frac{t_0}{\sqrt{2} \cdot s_{\mathrm{hi}}}\right)\right)}$$

where t_0 is the moment of the heavy ion penetration (see the keyword `Time` in Table 294 on page 1814), and s_{hi} is the characteristic value of the Gaussian (see `s_hi` in Table 145 on page 866).

The spatial distribution, $R(w, l)$, can be defined as either an exponential function (default):

Equation 821

$$R(w, l) = \exp\left(-\frac{w}{w_{\mathrm{t}}(l)}\right)$$

or a Gaussian function:

Equation 822

$$R(w, l) = \exp\left(-\left(\frac{w}{w_t(l)}\right)^2\right)$$

where w is a radius defined as the perpendicular distance from the track. The characteristic distance w_t is defined as `Wt_hi` in the `HeavyIon` statement and can be a function of the length l (see Table 294).

In addition, the spatial distribution $R(w, l)$ can be defined as a PMI function (see Heavy Ion Spatial Distribution on page 1604).

The LET generation density, $G_{\mathrm{LET}}(l)$, is given by:

Equation 823

$$G_{\mathrm{LET}}(l) = a_1 + a_2 l + a_3 e^{a_4 l} + k'\left[c_1(c_2 + c_3 l)^{c_4} + \mathrm{LET_f}(l)\right]$$

where $\mathrm{LET_f}(l)$ (defined by the keyword `LET_f`) is a function of the length l. Example 2 on page 867 illustrates how to use an array of values to specify the length dependence of $\mathrm{LET_f}(l)$. A linear interpolation is used for values between the array entries of `LET_f`.

There are two options for the units of `LET_f`: pairs/cm^3 (default) or pC/µm (activated by the keyword `PicoCoulomb`). Depending on the units chosen, k' takes on different values in order to make Equation 823 dimensionally consistent. Table 144 summarizes the appropriate values of k' for different device dimensions.

Exercise care when choosing the correct units for `Wt_hi` and `Length`. The default unit of `Wt_hi` and `Length` is centimeter. If you specify `PicoCoulomb`, then the unit becomes µm. For examples illustrating the correct unit use, see Examples: Heavy Ions on page 866.

Table 145 lists the other coefficients used in Equation 823 with their default values, and they can be adjusted in the parameter file of Sentaurus Device.

Note:

The keyword `Location` defines a bidirectional track for which Sentaurus Device computes the generation rate in both directions from the place of incidence along the `Direction` vector. The keyword `StartPoint` defines a one-directional track. In this case, Sentaurus Device computes the generation rate only in the positive direction from the place of incidence.

Table 144 Setting correct k' to make LET generation density equation dimensionally consistent

Condition	Two-dimensional device	Three-dimensional device
`LET_f` has units of pairs/cm^3 and $R(w, l)$ is exponential or Gaussian	$k' = k$	$k' = k$
`LET_f` has units of pC/µm and $R(w, l)$ is exponential	$k' = \frac{k}{2w_t d}$ $d = 1\,\mu\,\mathrm{m}$	$k' = \frac{k}{2\pi w_t^2}$
`LET_f` has units of pC/µm and $R(w, l)$ is Gaussian	$k' = \frac{k}{\sqrt{\pi}\, w_t d}$ $d = 1\,\mu\,\mathrm{m}$	$k' = \frac{k}{\pi w_t^2}$

Table 145 Coefficients for carrier generation by heavy ion (HeavyIon parameter set)

Coefficient	Keyword	Default value	Default unit	Unit if PicoCoulomb is chosen
s_{hi}	`s_hi`	2e-12	s	s
a_1	`a_1`	0	pairs/cm^3	pairs/cm^3
a_2	`a_2`	0	pairs/cm^3/cm	pairs/cm^3/µm
a_3	`a_3`	0	pairs/cm^3	pairs/cm^3
a_4	`a_4`	0	cm^{-1}	µm^{-1}
k	`k_hi`	1	1	1
c_1	`c_1`	0	pairs/cm^3	pC/µm
c_2	`c_2`	1	1	1
c_3	`c_3`	0	cm^{-1}	µm^{-1}
c_4	`c_4`	1	1	1

Examples: Heavy Ions

This section presents examples of using the heavy ion model.

Example 1

The track has a constant `LET_f` value of 0.2 pC/µm across the track. The track length is 1 µm (l_{max}= 1 µm) and the heavy ion crosses the device at time 0.1 ps. The unit

of LET_f is pC/μm and the spatial distribution is Gaussian. Since PicoCoulomb is specified, the values of Length and Wt_hi are expressed in terms of μm. The keyword HeavyIonChargeDensity in the Plot section plots the charge density generated by the ion.

```
Physics {
   Recombination ( SRH(DopingDependence) )
   Mobility (DopingDependence Enormal HighFieldSaturation)
   HeavyIon (
      Direction=(0,1)
      Location=(1.5,0)
      Time=1.0e-13
      Length=1
      Wt_hi=3
      LET_f=0.2
      Gaussian
      PicoCoulomb
   )
}
Plot { eDensity hDensity ElectricField HeavyIonChargeDensity }
```

Example 2

The LET_f and radius (Wt_hi) values are functions of the position along the track (in this case, l_{max} = 1.7×10^{-4} cm). Values in between the array entries are interpolated linearly. The unit of LET_f is pairs/cm^3 (because PicoCoulomb is not specified), and the unit for Length and Wt_hi is centimeter. For each value of length, there is a corresponding value of LET_f and a value for the radius. The spatial distribution in the perpendicular direction from the track is exponential.

```
Physics {
   Recombination ( SRH(DopingDependence) )
   HeavyIon (
      Direction=(0,1)
      Location=(1.5,0)
      Time=1.0e-13
      Length = [1e-4 1.5e-4 1.6e-4 1.7e-4]
      LET_f = [1e6 2e6 3e6 4e6]
      Wt_hi = [0.3e-4 0.2e-4 0.25e-4 0.1e-4]
      Exponential
   )
}
```

Example 3

This example illustrates multiple ion strikes in the model.

```
Physics {
   Recombination ( SRH(DopingDependence) )
   HeavyIon (
      Direction=(0,1)
      Location=(0,0)
```

```
        Time=1.0e-13
        Length = [1e-4 1.5e-4 1.6e-4 1.7e-4]
        LET_f = [1e6 2e6 3e6 4e6]
        Wt_hi = [0.3e-4 0.2e-4 0.25e-4 0.1e-4]
    )
    HeavyIon ("Ion1")(
        Direction=(0,1)
        Location=(1,0)
        Time=1.0e-13
        Length = [1e-4 1.5e-4 1.6e-4 1.7e-4]
        LET_f = [1e6 2e6 3e6 4e6]
        Wt_hi = [0.3e-4 0.2e-4 0.25e-4 0.1e-4]
    )
}
```

User-defined heavy ions do not have default parameters. Therefore, in the parameter file, a specification of the following form must appear (see Table 145 on page 866):

```
HeavyIon("Ion1")(...)
```

References

[1] J.-L. Leray, “Total Dose Effects: Modeling for Present and Future,” *IEEE Nuclear and Space Radiation Effects Conference (NSREC) Short Course*, 1999.

[2] A. Erlebach, *Modellierung und Simulation strahlensensitiver Halbleiterbauelemente*, Aachen: Shaker, 1999.

[3] L. C. Northcliffe and R. F. Schilling, “Range and Stopping - Power Tables for Heavy Ions,” *Nuclear Data Tables*, vol. A7, no. 1–2, New York: Academic Press, pp. 233–463, 1969.